

Introduction to Single Cell RNA sequencing with 10X Genomics

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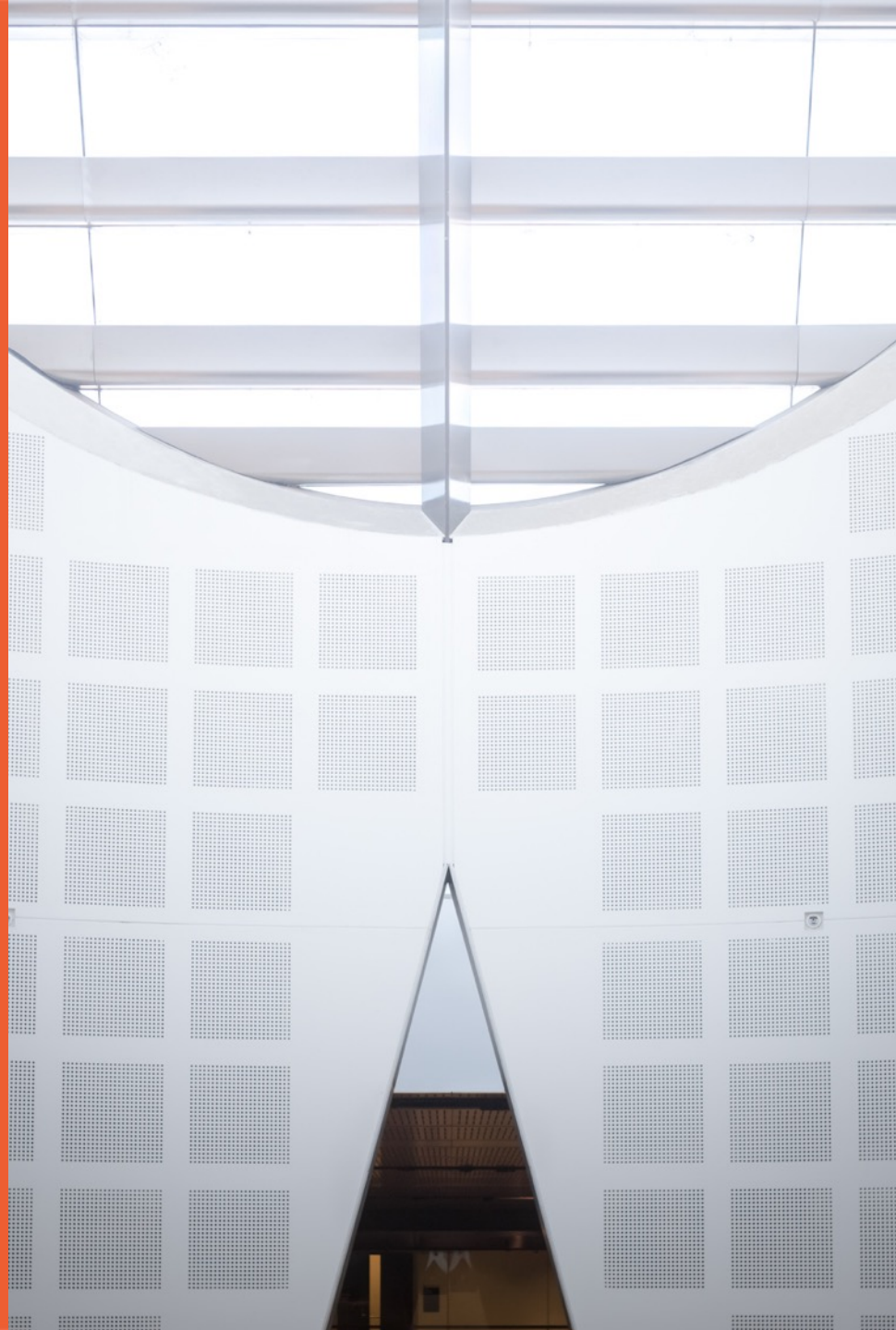
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Acknowledgements:

10X Genomics



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About this course

Introduction to:

- Artemis HPC
- Single Cell RNA sequencing with the 10X Chromium system
- 10X Genomics' bioinformatics pipelines Cell Ranger
- 10X Genomics' Loupe Cell browser

By the end of the course:

- You will be familiar with the compute resource Artemis HPC
- Understand how single cell RNA sequencing works using the 10X system
- Know how to run an end-end QC and analysis pipeline using cellranger
- Know how to visualize results using Loupe Cell Browser

Pre-requisites: Intro to RNA-Seq on Galaxy

Additional training: Intro to Artemis HPC/Data transfer and Research Data Storage (RDS) for HPC

Course outline

1. Artemis HPC (1.30pm – 2.30pm)

- Getting on to Artemis
- Artemis 101
- Downloading the data
- Submitting scripts

2. Introduction to Single Cell RNA sequencing (2.30pm – 2.50pm)

- Bulk RNA sequencing vs Single Cell RNA sequencing
- Single Cell v3 Chemistry & Chromium System (10X Genomics)
- Common workflows

3. Understanding the Data and using cellranger (2.50 – 3.10pm)

- Demultiplexing with “mkfastq”
- FASTQ, clustering and differential expression analysis with “count”

4. Cell Ranger output (3.10pm – 4.00pm)

- web_summary.html, quality checking
- Loupe Cell Browser for interactive display of results (cloupe.cloupe)

5. Additional pipelines and resources (4.10pm – 4.30pm)

- Additional pipelines: “aggr” and “re-analyze”
- Links to useful resources

Today's exercise

Dataset 1

Sample: Peripheral blood mononuclear cells (PBMCs) from a healthy donor

Chemistry: Chromium Single Cell 3' v3 chemistry

Sequencing: Illumina NovaSeq

Source: Publicly available, provided by 10X Genomics

Process & analyse raw sequencing data using cellranger count on Artemis.

- QC metrics
- Alignment, cell and UMI counting
- Unsupervised clustering (Graph-based, K-means)
- Differential expression between clusters

You can go through the results from this dataset in your own time.

Today we will go through the results of a different dataset.

Today's exercise

Dataset 2

Sample: Peripheral blood mononuclear cells (PBMCs) from a healthy donor

Chemistry: Chromium Single Cell 3' v3 chemistry

Sequencing: Illumina NovaSeq

Source: Publicly available, provided by 10X Genomics

Visualise output from cellranger count using 10X Genomics' software Loupe Cell Browser:

- QC metrics in 'web_summary.html'
- Visualise unsupervised clustering results (t-SNE plots)
- Identify cell types using known markers

Section 1: Artemis HPC



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1. Artemis HPC

What is Artemis HPC?

HPC stands for ‘High Performance Computing’, but you might also simply call Artemis a ‘supercomputer’. Technically, Artemis is a *computing cluster*, which is a whole lot of individual computers networked together. At present, Artemis consists of:

- 7,636 cores (CPUs)
- 45 TB of RAM
- 108 NVIDIA V100 GPUs
- 378 TB of storage
- 56 Gbps FDR InfiniBand (networking)

Artemis computers use a **Linux** operating system, ‘CentOS’ v6.9. Computing performed on Artemis nodes is managed by a *scheduler*, and ours is an instance of ‘PBS Pro’.

1. Why do we need to use Artemis?

cellranger system requirements

Local Mode

- Run on single, standalone Linux system
- 64-bit CentOS/RedHat 6.0 or Ubuntu 12.04
- 8 cores, 64GB RAM (minimum)

Runtime

- 7865 cells, 35433 reads/cell
- ~8 hours wall-time at minimum compute resources

- The pipeline you will run is an end-end clustering analysis using cellranger
- This pipeline also produces input files to other popular open source software (e.g. Seurat, Monocle2...)

1. Set up

To use Artemis, you will need to a shell terminal emulator

Please follow the instructions on this page:

<https://sydney-informatics-hub.github.io/training.artemis.introhpc/setup.html>

- Mac users: terminal, iTerm2
- Windows users: X-Win32, Putty

Today we will assign training unikeys for you to use in this course.

The training unikey is:

ict_hpctrainN

(N = 1– 40, we will assign you a number)

1. Artemis 101

Once you've successfully logged onto Artemis, you should see something like this on your terminal screen:

```
-----
***** ARTEMIS 3.0 *****
-----
***   !!! LOGIN NODES ARE FOR JOB SETUP & SUBMISSION ONLY !!!   ***
-----
Use Interactive flag (-I) or defaultQ or small-express queue for job testing

NO ACTUAL COMPUTE WORK IS TO BE CONDUCTED ON LOGIN NODES

Any sustained compute or multicore/memory/io intensive processes on login nodes
will be subject to being killed without notice.
-----
!!! MEMORY MUST BE SPECIFIED FOR ALL JOBS - use the mem= PBS directive !!!
-----
Please refer to the Quick Start User Guide to get started with Artemis 2.0:
      https://informatics.sydney.edu.au/services/artemis/
-----

*****
Artemis will be offline for maintenance from 6am June 24 to 11:59pm June 25.
Jobs scheduled to run during this window will remain queued until maintenance
completes.
*****
!! Connected to Training instance.
[TRAINING ict_hpctrain1@login4 ~]$
```

Artemis is a computer (just like your Mac or Windows PC) except that you interact with it via the **command line** instead of using your mouse to point and click (graphical user interface – GUI).

1. Artemis 101

At the command prompt, e.g. after [TRAINING ict_hpctrain1@login4 ~]\$ you can interact with Artemis by typing in commands. Commands can be followed by options

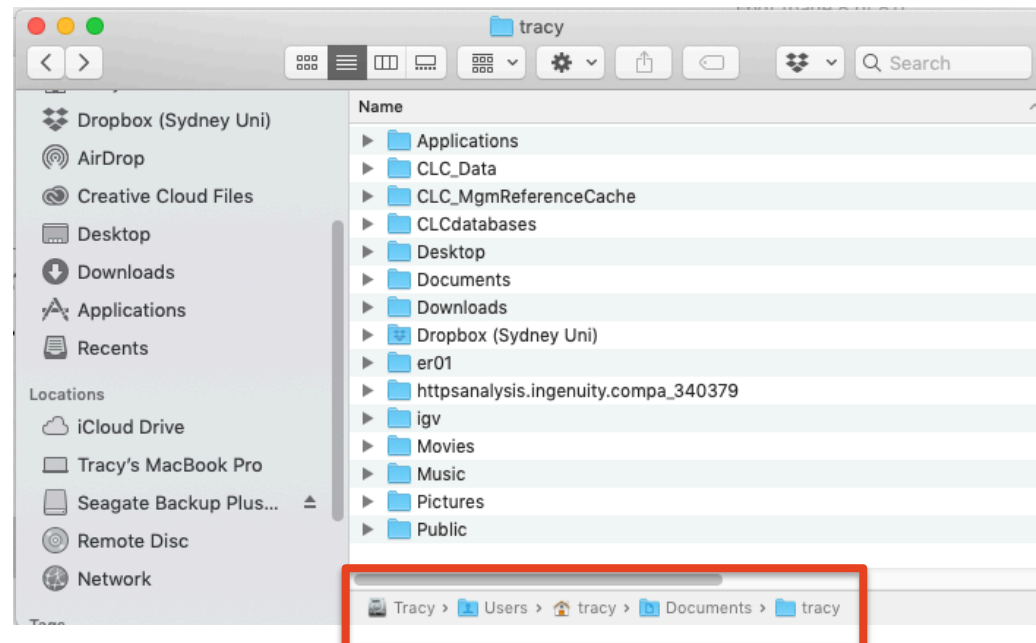
Type in your first command:

```
pwd
```

This command shows the “path to working directory”. It shows which folder/subfolders we are in.

Which folder/subfolders are you in when logging onto Artemis?

On a mac:



1. Artemis 101

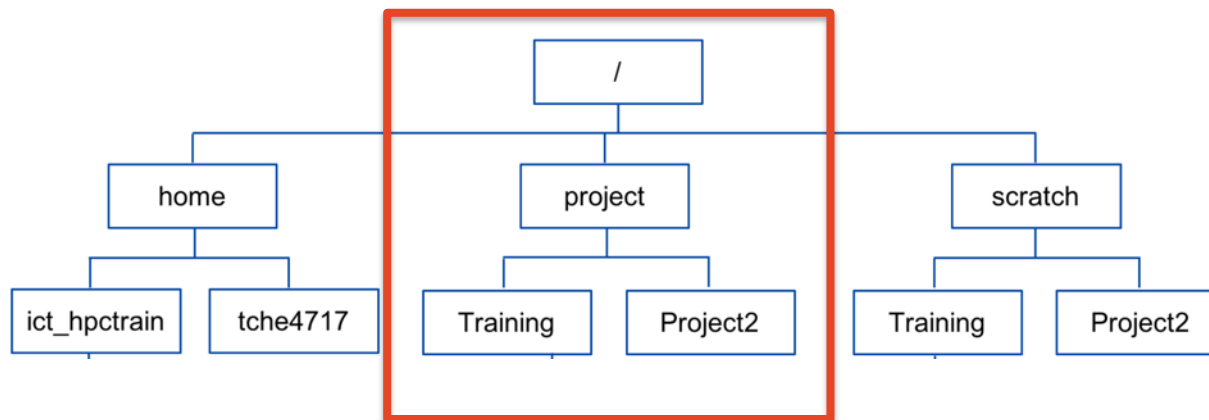
There is only 10GB of space in the home directory.

Change to the Training folder within the project directory (1Tb):

```
cd /project/Training
```

Create your own directory (replace YourName) to work in for today:

```
mkdir YourName
```



1. Downloading the data

Go into the directory that you just created:

```
cd YourName
```

Download today's data (please type the command, copy the URL)

```
wget -O SC_workshop.tar.gz <url>
```

Replace <url> with: <https://cloudstor.aarnet.edu.au/plus/s/vu0XtM4V408DGgg/download>

Unpack the data:

```
tar -zxvf SC_workshop.tar.gz
```

Enter the SC_workshop directory

```
cd SC_workshop
```

Take a look at the directories and files that you have downloaded:

```
tree
```

1. Submitting scripts

Scripts are a **set of commands** that is interpreted by the computer and executed in sequence.

On Artemis, we need to submit jobs through a **PBS submission script**.

```
2. jdar4135@login2:/project/Training/hayim (ssh)
GNU nano 2.0.9 File: basic.pbs

#!/bin/bash ← the 'shebang'
#Create indexes for reference sequence
#PBS -P Project
#PBS -N Index_YourName
#PBS -l select=1:ncpus=1:mem=4GB
#PBS -l walltime=00:10:00
#PBS -q small-express

# Load modules
module load bwa/0.7.17
module load samtools/1.6

io=/project/Project/YourName

# BWA index:
bwa index -a bwtsv ${io}/canfam3_chr5.fasta

# SAMtools index:
samtools faidx ${io}/canfam3_chr5.fasta

^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell
```

1. Submitting scripts

We will edit and submit the `cellranger_count.pbs` script
(we will go through what this does in depth in section 2 of the course)

Go to directory that `cellranger_count.pbs` is in:

```
cd Scripts
```

You need to use a text editor, such as `nano`, to edit the script:

```
nano cellranger_count.pbs
```

* Tip: options in `nano` are provided at the bottom of the screen. Press `ctrl` in place of the “`^`” character

1. Submitting scripts

Can you identify:

- `Shebang` line
- PBS directives
- Module loads*
- Job commands

Edit the 'mydir' and 'email' variables

To save:

- ctrl + o, enter

To exit:

- ctrl + x

```
GNU nano 2.0.9 File: cellranger_count.pbs
#!/bin/bash

#-----
# Software: cellranger/3.0.2
# Platform: University of Sydney's Artemis HPC
# Description: Run cellranger count
#
# Author: Tracy Chew (tracy.chew@sydney.edu.au)
# Date last modified: 27/06/2019
#
# If you use Artemis or this script towards a publication, please
# acknowledge the Sydney Informatics Hub (or include the author of this
# script co-authorship, when appropriate).
#
# Suggested acknowledgement:
# This research was supported by the Sydney Informatics Hub and the
# University of Sydney's high performance computing service, Artemis
#-----

#PBS -P Training
#PBS -N cellranger
#PBS -l select=1:ncpus=4:mem=24GB
#PBS -l walltime=10:00:00
#PBS -q defaultQ

# Change the 'mydir' variable to your own directory
# Change the 'email' variable to your own email
mydir=YourName
email=Your.Email@sydney.edu.au

# You do not need to change anything else in this script
# These variables do not need to be changed
project=Training
ref=/project/Training/SingleCell/refdata-cellranger-GRCh38-3.0.0
dir=/project/Training/${mydir}/SC_workshop
fasta=${dir}/pbmc_1k_v3_fastqs
sample=pbmc_1k_v3
id=pbmc_1k_v3

# Make a directory called CellRanger and change into it. The output will be stored here.
mkdir -p ${dir}/CellRanger
cd ${dir}/CellRanger

# Run cellranger count
echo "$(date): Running cellranger count"
cellranger count \
    --localcores=${NCPUS} \
    --localmem=24 \
    --transcriptome=${ref} \
    --id=${id} \
    --fastqs=${fasta} \
    --sample=${sample}

cd ${dir}/outs

# Email the results to yourself. You should receive these after the course is complete
# Go through the results of this analysis in your own time
# NOTE: Emails only work on the training node.
# Please attend Intro to Artemis and Data transfer and Research Data Storage (RDS) for HPC to learn how to
# transfer data on Artemis
echo "$(date): Mailing web_summary.html and cloupe.cloupe results"
echo "Results from today's single cell course" | mutt \
    -s "Single Cell Results" \
    -a web_summary.html \
    -a cloupe.cloupe -- \
    ${email}
```

AG Get Help AR WriteOut AY Read File AY Prev Page AX Cut Text AC Cur Pos
AX Exit AJ Justify AW Where Is AV Next Page AU UnCut Text AT To Spell

1. Submitting scripts

When you submit a job on Artemis, 3 log files will be generated at completion of the job and saved to where you submitted the job.

To keep things tidy, let's submit the job in a "Logs" directory.

First, create the directory:

```
mkdir Logs
```

Submit the job:

```
qsub ../cellranger_count.pbs
```

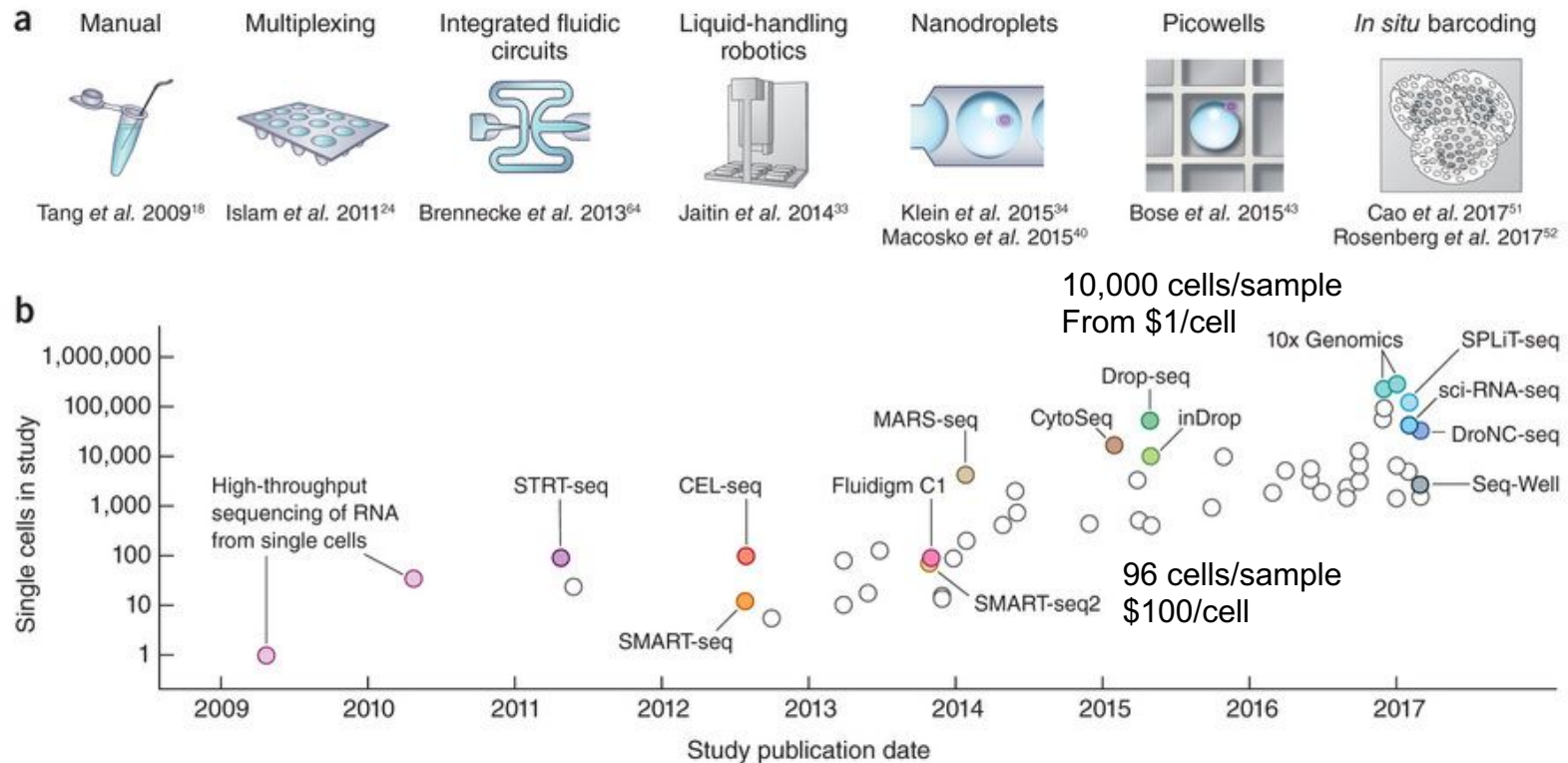
Check the status of your job:

```
qstat -u ict_hpctrainN
```

Section 2: Single Cell RNA sequencing with 10X Chromium



2. Single-cell protocols



2. Bulk RNA-Seq vs Single Cell RNA-Seq

Bulk RNA-sequencing

Average expression level for each gene across a large population of input cells

Single-cell RNA-sequencing (scRNA-seq)

Gene expression profile of individual cells are measured

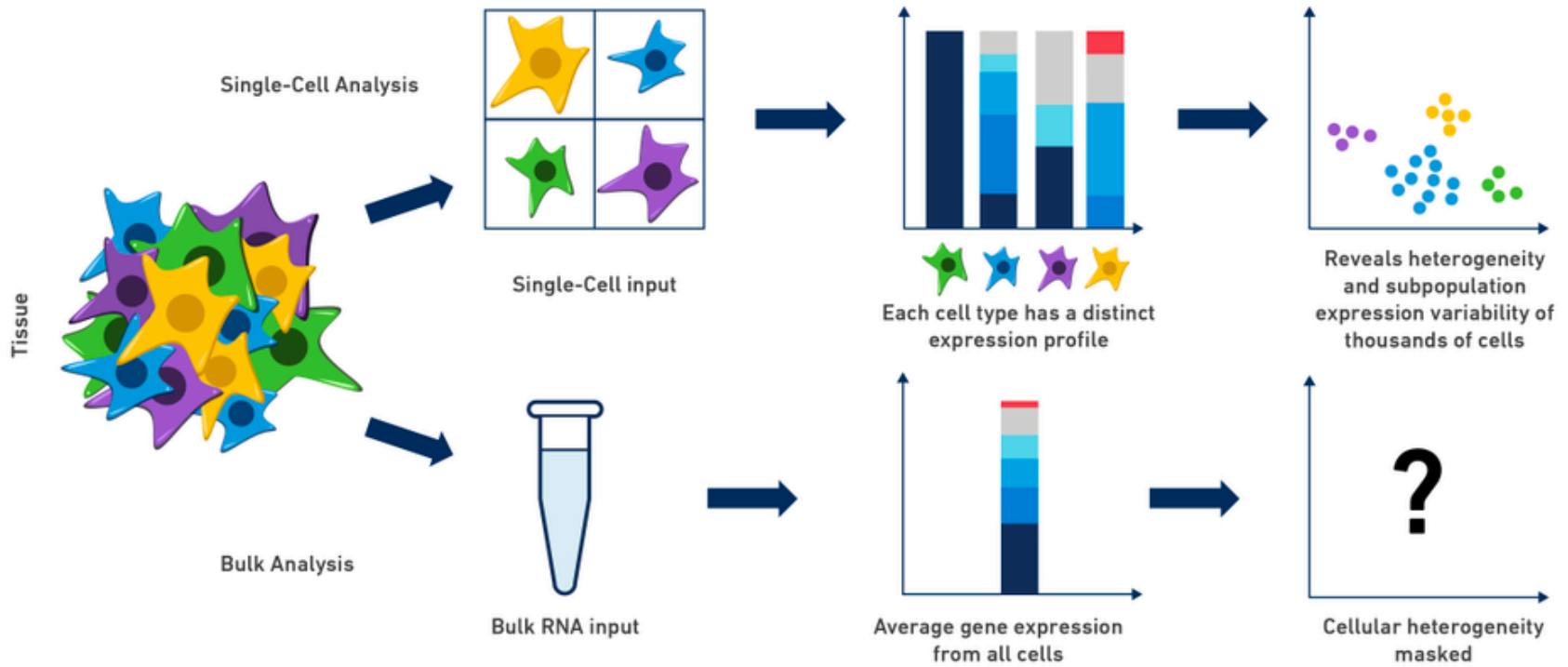
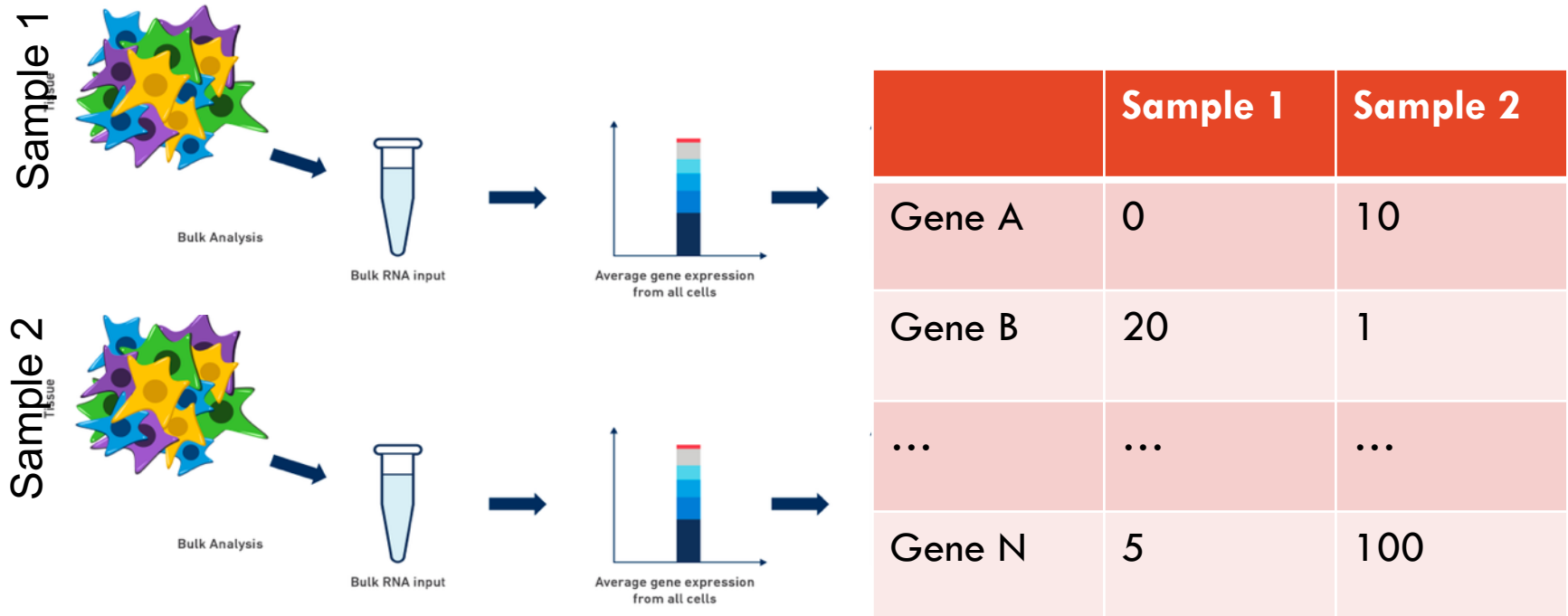


Image: 10X Genomics

2. Gene expression is measured by “counts”

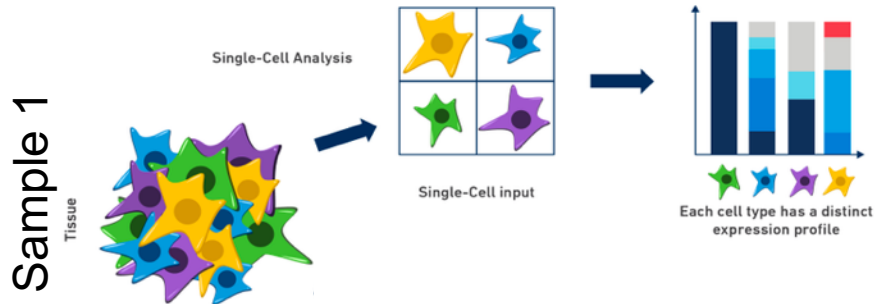
Bulk RNA sequencing



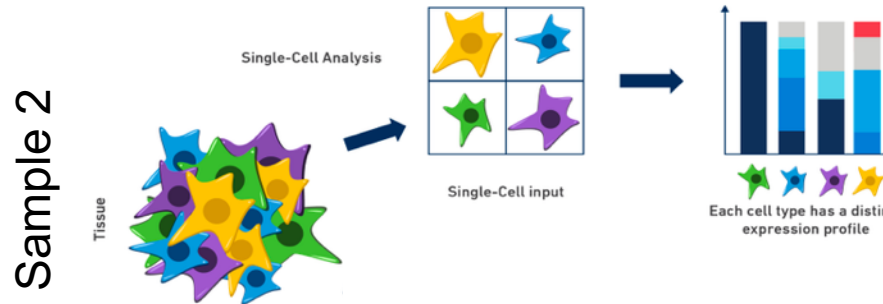
Sample > Library Prep > Sequencing > Raw reads > Alignment > Count matrix

2. Gene expression is measured by “counts”

Single-Cell RNA sequencing



	Cell 1	Cell 2
Gene A	0	10
Gene B	20	1
...	...	...
Gene N	5	100



	Cell 1	Cell 2
Gene A	0	10
Gene B	20	1
...	...	...
Gene N	5	100

- Per sample (sample barcoding)
- Per cell (10X cell barcoding)
- Per transcript (Unique molecular identifier - UMI)

2. 10X Genomics Chromium Workflow

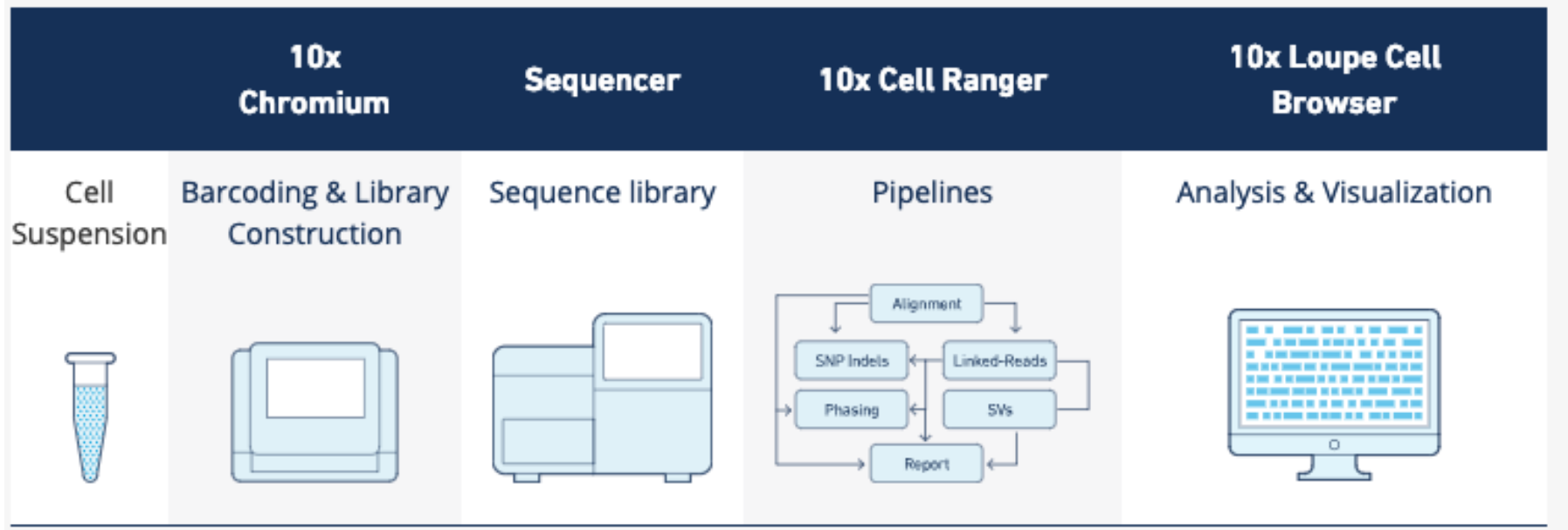
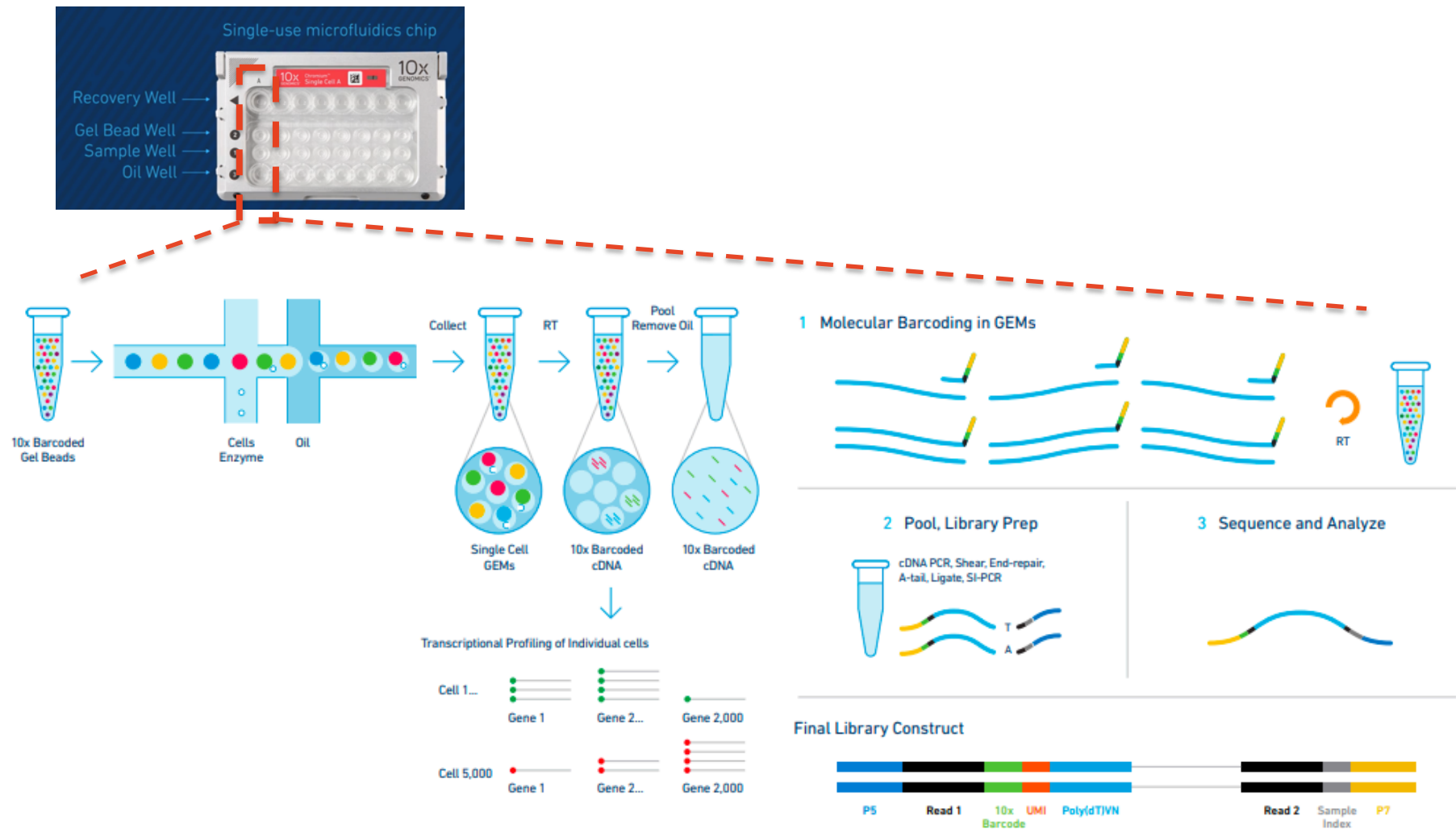


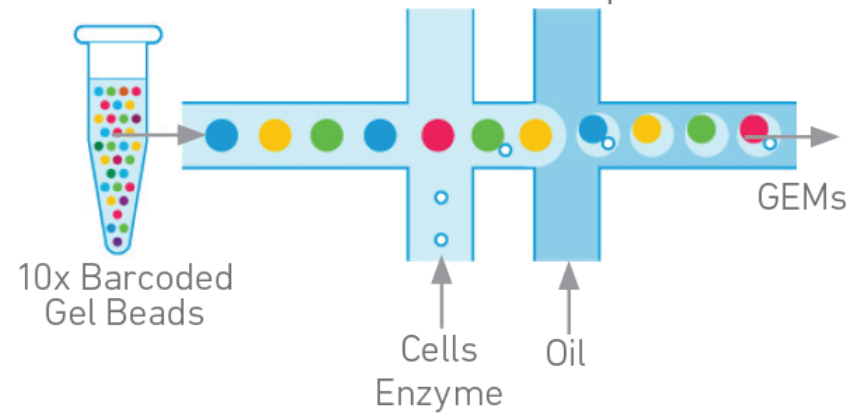
Image: 10X Genomics

2. 10X Genomics Chromium Workflow



2. 10X Genomics Chromium

- Microdroplet based method
- 8 channels processed in parallel
- 500 – 10,000 cells per channel
- 30µm cell diameter
- 3'-cDNA library (tag-based)
- Run takes < 7 minutes.
- Recovers up to ~65% of cells (average 50%)
- Requires >80% viable cells
- Low doublet rate (~3.9% per 5,000 cells)
- Compatible with popular Illumina platforms (HiSeq, MiSeq, NextSeq, NovaSeq)
- For the most applications an average of 20,000 reads per cell should be sequenced (for cell types with complex transcriptomes, v3 chemistry)



2. Overview

Raw Illumina BCLs

FASTQ

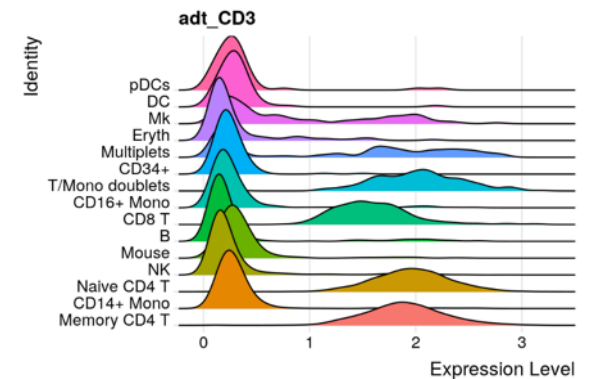
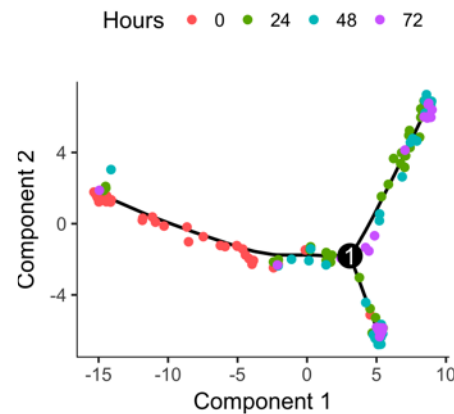
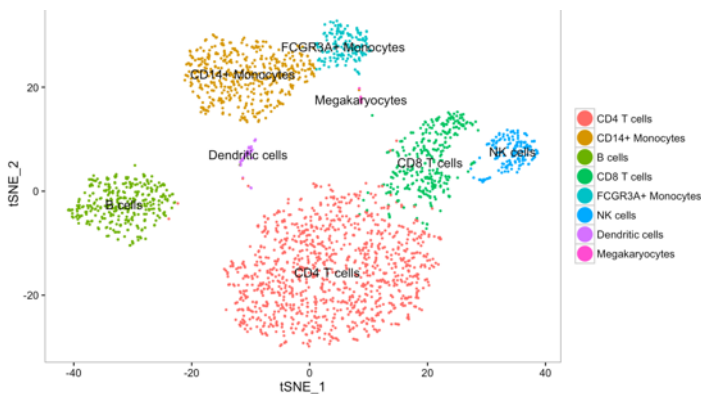
Alignment, cell barcode and
UMI counting

Feature-barcode matrices

Cell type identification

Cell Trajectory Analysis

Observe protein levels on
single cell clusters



2. Overview

>90 tools
currently
available

Raw Illumina BCLs

FASTQ

Alignment, cell barcode and
UMI counting

Cell Ranger
Partek

Feature-barcode matrices

Cell type identification

Cell Trajectory Analysis

Observe protein levels on
single cell clusters

Cell Ranger
Seurat
SC3
scrn
Partek

TSCAN
Monocle 2
DDRTree

Cell Ranger
Seurat

2. Clustering tools to analyse 10X data

RESEARCH ARTICLE

REVISED Comparison of clustering tools in R for medium-sized 10x Genomics single-cell RNA-sequencing data [version 2; referees: 3 approved]

Saskia Freytag ^{1,2}, Luyi Tian ^{2,3}, Ingrid Lönnstedt⁴, Milica Ng⁴,
Melanie Bahlo ^{1,2}

¹Population Health and Immunity, V

²Department of Medical Biology, U

³Molecular Medicine Division, W

⁴Bio21 Institute, CSL Limited, Park

Table 1. Overview of the clustering tools included in this study, and several characteristics thereof.

Software	Year	Similarity Metric	Clustering Method	Ref
ascend	2017	Euclidean distance	Hierarchical clustering	7
Cell Ranger	2016	Euclidean distance	Graph-based clustering	
CIDR	2017	Imputed dissimilarity	Hierarchical clustering	8
countClust	2014	none	Grade of membership models	9
RaceID	2015	Pearson correlation	K-means clustering	10
RaceID2	2016	Pearson correlation	K-means clustering	11
RCA	2017	Pearson correlation	Supervised hierarchical clustering	12
SC3	2016	Euclidean distance	Consensus clustering	13
scrn	2016	Euclidean distance	Hierarchical clustering	14
Seurat	2015	Euclidean distance	Graph-based clustering	15
SIMLR	2016	Multikernel learning	Spectral clustering	16
TSCAN	2016	none	Model-based clustering	17

Table 2. Properties of all benchmarking datasets used in the study.

Benchmark standard	Gold	Silver				
Dataset		Dataset 1	Dataset 2/2a	Dataset 3/3a	Dataset 4	Dataset5
Tissue	Cell lines	PBMCs	PBMCs	PBMCs	PBMCs	PBMCs
Source	GSE111108	GSE115189	Website ^a / ^b	Website ^a / ^b	Website ^a	Website ^b
Instrument	Chromium	Chromium	GemCode	Chromium	GemCode	Chromium

2. Clustering tools to analyse 10X data

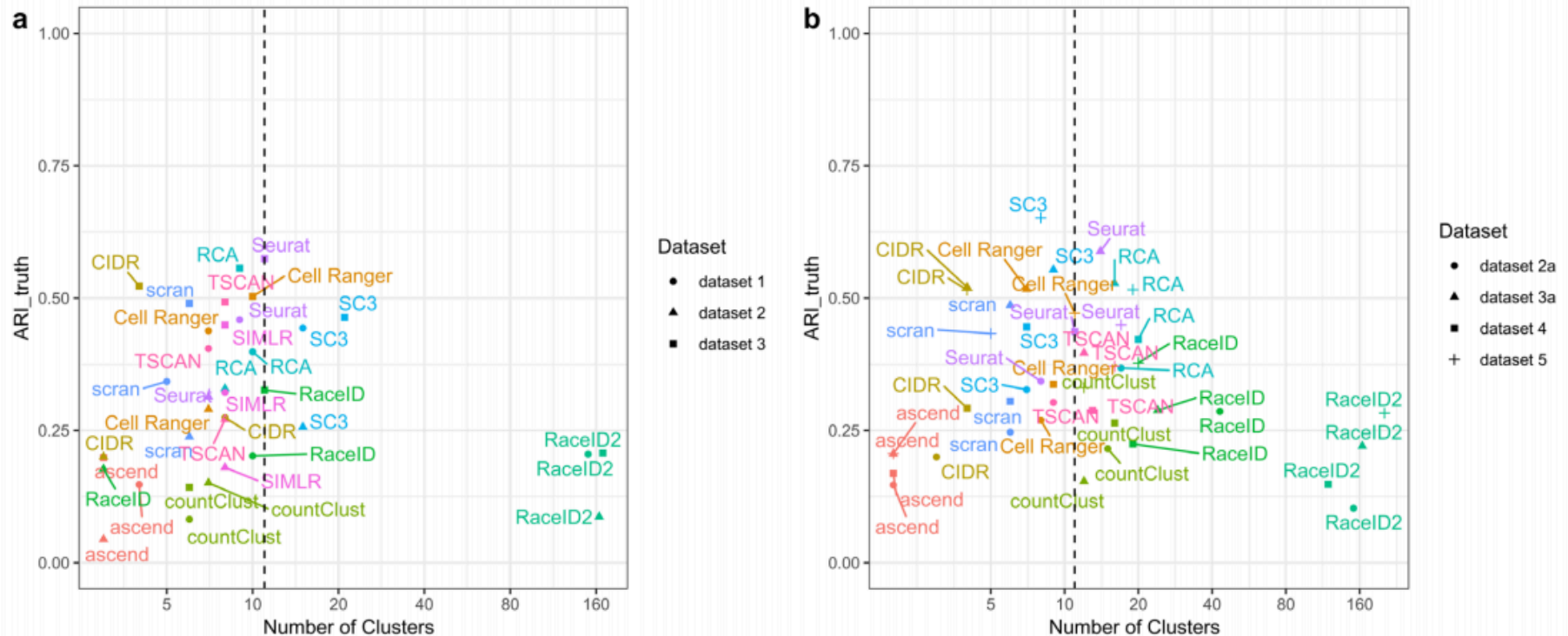


Figure 2. ARI_truth of each method in each dataset, as indicated by different shapes, with regards to the supervised cell labeling versus the number of clusters. The dashed line indicates the number of cell populations estimated by the supervised cell labeling approach. (a) First evaluation with methods available in R 3.4.3. (b) Second evaluation with methods available in R 3.5.0.

Figure: Freytag et al, 2018

2. Selecting the best methods...

Only Seurat, SC3 and Cell Ranger achieved an ARI_truth above 0.4 for at least two datasets in each of the evaluations. All

While Seurat performed slightly better than the next best methods, in our opinion, the choice of clustering method should be informed by the user's familiarity with statistical concepts and R programming. Many methods, including Seurat, require the user to make informed parameter choices and occasionally troubleshoot code. Methods requiring no parameter choices, like Cell Ranger, may offer a better choice for non-experts.

Section 3: 10X Genomics' Cell Ranger pipeline



3. 10X Genomics' CellRanger

Cell Ranger

1. cellranger mkfastq
 - Creates fastq from BCL and some quality reports
2. cellranger count
 - Performs alignment (STAR), filtering, barcode counting, and UMI counting **for one sample**
 - Chromium cellular barcodes are used to generate gene-barcode matrices, determine clusters and perform gene expression analyses
3. [optional] cellranger aggr
 - Aggregates output from cellranger count **from multiple samples**
 - Normalizes the combined data according to sequencing depth
4. [optional] cellranger reanalyze

Loupe Cell Browser

Interactive display of t-SNE plots and differential expression genes

3. cellranger mkfastq

cellranger mkfastq demultiplexes Illumina BCL files into standard FASTQ files.

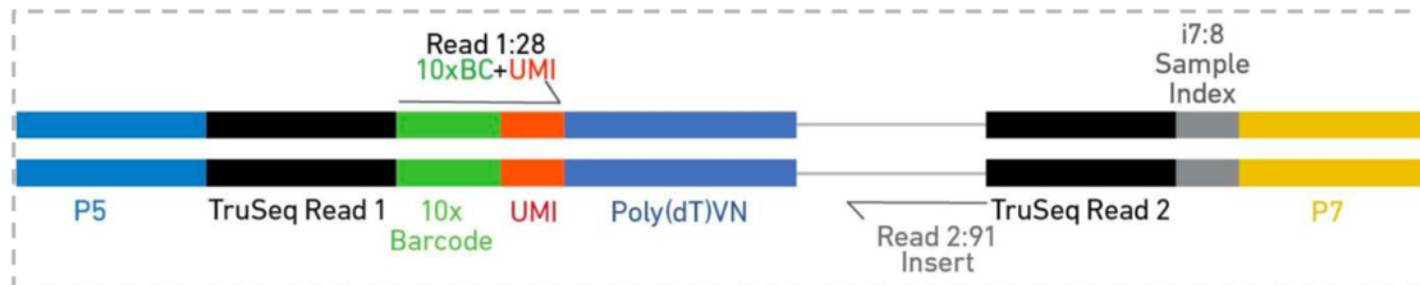
This is a wrapper for Illumina's bcl2fastq tool

**** Our files have already been demultiplexed, so you can skip this step ****

```
cellranger mkfastq \  
  --id=sample1 \  
  --csv=my_samples.csv \  
  --run=/mnt/hiseq/sample1_bcl
```

3. cellranger mkfastq

Recall from the Single Cell 3' Gene Expression Library:



Minimum 20,000 read pairs per cell is recommended for 3' Gene Expression libraries

	Read 1	i7 Index	i5 Index	Read 2
Purpose	10x Barcode & UMI	Sample Index	N/A	Insert (Transcript)
Length	28* (16 + 12)	8	0	91**

3. cellranger mkfastq

Take a look at your fastq files:

```
cd /project/Training/YourName/SC_workshop/pbmc_1k_v3_fastqs
```

FASTQ files are the standard file format for raw sequencing data.

```
[TRAINING ict_hpctrain1@login3 pbmc_1k_v3_fastqs]$ zcat pbmc_1k_v3_S1_L001_R1_001.fastq.gz | head -4
@A00228:279:HFWFVDMXX:1:1101:8486:1000 1:N:0:NCATTACT
NGTGATTAGCTGTACTCGTATGTAAGGT
+
#FFFFFFFFFFFFFFFFFFFFFFFF
```

See [Wikipedia](https://en.wikipedia.org/wiki/FASTQ_format) for a good description of the FASTQ file format.

- How many fastq files do you see?
- Can you identify:
 - The sample index
 - The cell index (10X barcode)
 - UMI (tags unique transcripts)

3. cellranger count

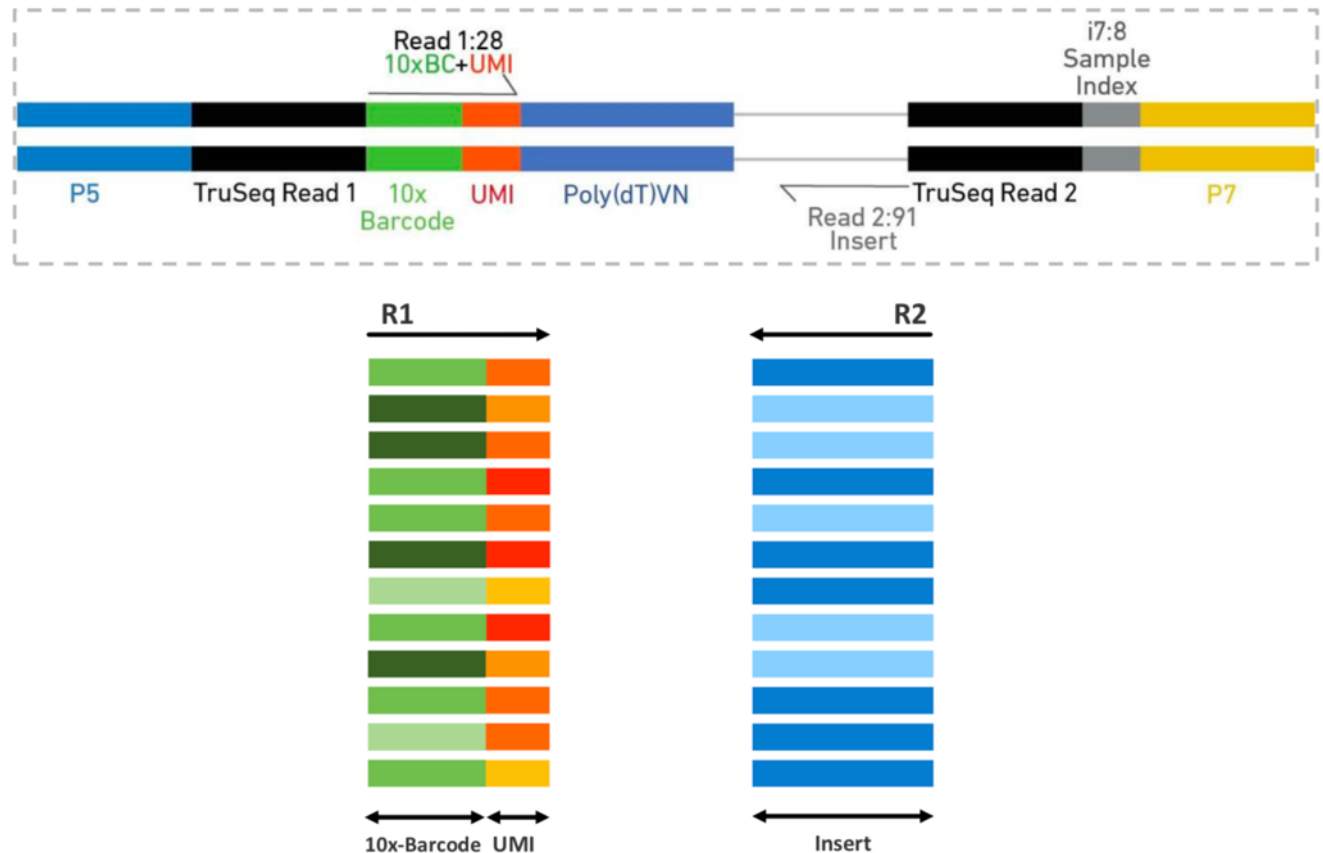
cellranger count performs read alignment, UMI counting, cell calling and secondary analysis for one sample.

**** This is what we ran today ****

```
cellranger count \  
  --localcores=${NCPUS} \  
  --localmem=24 \  
  --transcriptome=${ref} \  
  --id=${id} \  
  --fastqs=${fastq} \  
  --sample=${sample}
```

3. cellranger count

Extract Barcode, UMI
Align
Correct Barcodes
Filter UMIs
Count UMIs
Call Cells
Secondary Analysis



3. Reference

10X uses ENSEMBL genomes and gene annotations.

```
# You do not need to change anything else in this script
# These variables do not need to be changed
project=Training
ref=/project/Training/SingleCell/refdata-cellranger-GRCh38-3.0.0
dir=/project/Training/${mydir}/SC_workshop
fastq=${dir}/pbmc_1k_v3_fastqs
sample=pbmc_1k_v3
id=pbmc_1k_v3
```

10X Genomics provides pre-built references:

- Human hg19/GRCh37 and hg38/GRCh38
- Mouse (GRCm38)
- Human + mouse (GRCh38 + GRCm38)
- ERCC

You can build your own reference:

- Use 10X's mkref tool
- FASTA and GTF file
- Supports any reference compatible with STAR

3. cellranger count

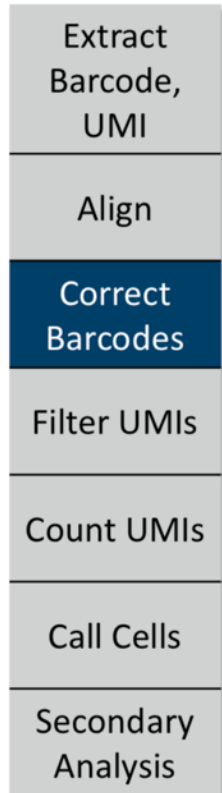
Extract Barcode, UMI
Align
Correct Barcodes
Filter UMIs
Count UMIs
Call Cells
Secondary Analysis



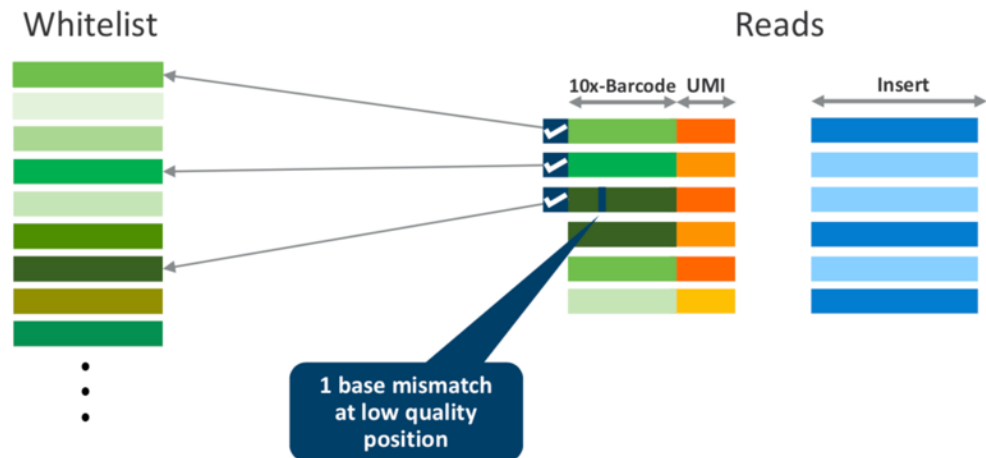
Alignment done via STAR (Spliced Transcripts Alignment to a Reference)

- Robust, open-source, junction-aware RNA-seq aligner
- Aligns reads to the genome and transcriptome simultaneously

3. cellranger count



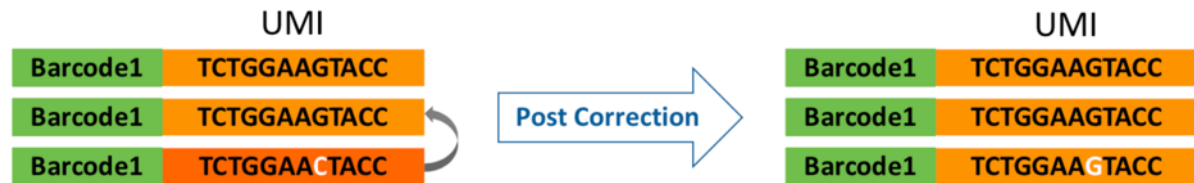
- Cell barcodes
 - Must be on a list of known cell barcode sequences
 - May allow 1 bp mismatch if the mismatch occurs at a low-quality position (the barcode is then corrected).



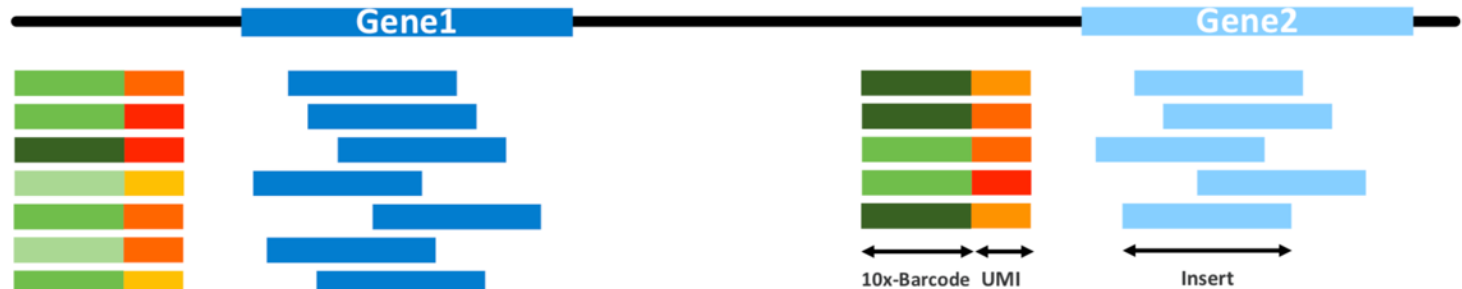
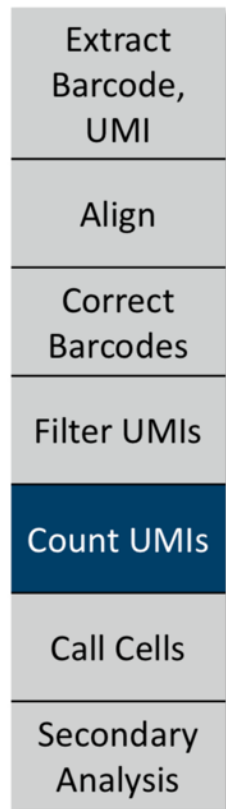
3. cellranger count

Extract Barcode, UMI
Align
Correct Barcodes
Filter UMIs
Count UMIs
Call Cells
Secondary Analysis

- Filter UMIs
 - Must not be a homopolymer, e.g. AAAAAAAAAA
 - Must not contain N
 - Must not contain bases with base quality < 10
- Correct UMIs
 - UMIs that are 1 mismatch away from a higher count UMI are corrected to that UMI if they share a cell barcode and gene.



3. cellranger count



Raw feature barcode matrix

	Genes	
	Gene1	Gene2
Barcode1	3	2
Barcode2	2	0
Barcode3	1	2

- Only the confidently mapped reads with valid barcodes and UMIs are used for counting
- Counting is done for all observed barcodes generating an unfiltered feature barcode matrix

3. cellranger count

Extract
Barcode,
UMI

Align

Correct
Barcodes

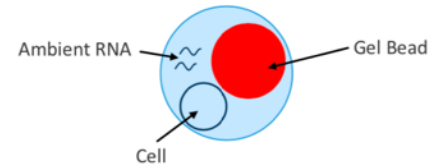
Filter UMIs

Count UMIs

Call Cells

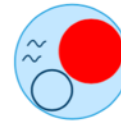
Secondary
Analysis

- One Chromium Single Cell Chip A or Chip B channel generates approximately 100,000 GEMs



- What is in the ~100,000 GEMs?

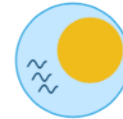
- A fraction of GEMs contains 1 cell each



- Typically very small number (depending of cell load) contain more than one cells



- A large fraction of GEMs are considered “empty,” without any cells, but do contain ambient RNA. These are called background barcodes.



- Need to distinguish GEMs/barcodes containing cells from background barcodes

3. cellranger count

- Secondary analysis done only on cells

Filtered Feature Barcode Matrix

Genes	Cell barcodes			
	1	2	3	4
0	0	0	0	0
9	3	0	0	0
0	0	10	0	0
0	0	5	0	0

PCA

t-SNE

Clustering

• Graph-based
• K-Means

Differential
Expression

sSeq



3. cellranger count output

- QC metrics: **web_summary.html**, metrics_summary.csv
- Feature barcode matrices (MEX and HDF5 formats)
 - Input to other popular software (e.g. Seurat, Monocle2)
- Position-sorted and indexed BAM file
- Secondary Analysis
 - CSV file with PCA analysis
 - CSV file with t-SNE projections
 - Cluster assignments for each cell are in clusters.csv for both K-means and graph based clustering
 - CSV file indicating which genes are differentially expressed in each cluster relative to all other clusters
- **.cloupe** file for visualization in Loupe Cell Browser

Section 4: cellranger count output



4. cellranger count output

Dataset 2

Sample: Peripheral blood mononuclear cells (PBMCs) from a healthy donor

Chemistry: Chromium Single Cell 3' v3 chemistry

Sequencing: Illumina NovaSeq

Source: Publicly available, provided by 10X Genomics

Visualise output from cellranger count using 10X Genomics' software Loupe Cell Browser:

- QC metrics in 'web_summary.html'
- Visualise unsupervised clustering results
- Identify cell types using known markers

Summary:

http://cf.10xgenomics.com/samples/cell-exp/3.0.0/pbmc_10k_v3/pbmc_10k_v3_web_summary.html

Loupe Cell Browser:

http://cf.10xgenomics.com/samples/cell-exp/3.0.0/pbmc_10k_v3/pbmc_10k_v3_cloupe.cloupe

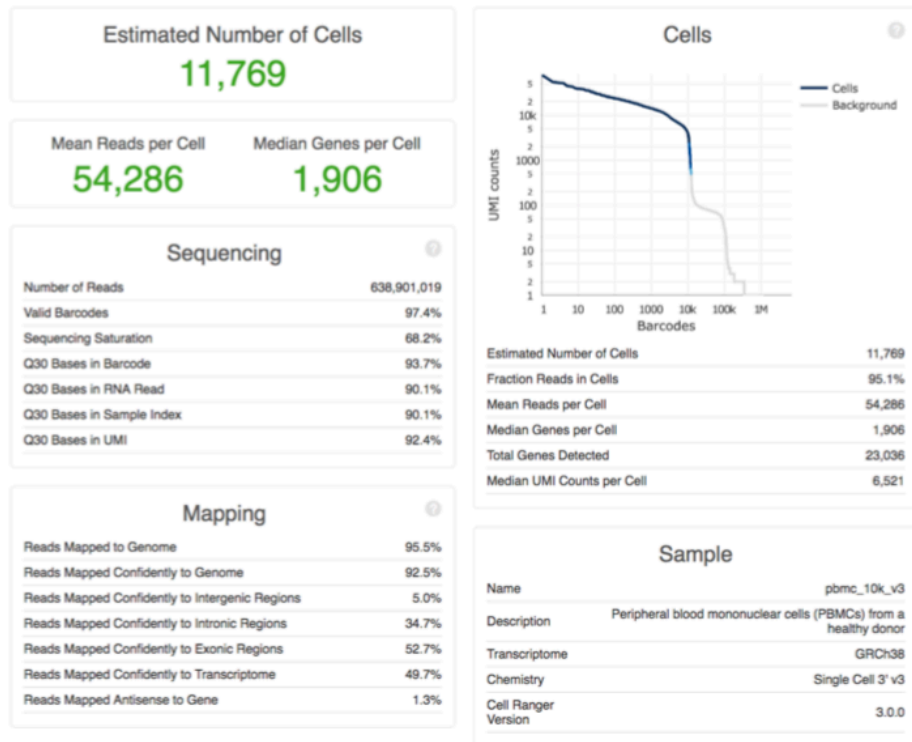
Known PBMC markers:

<https://support.10xgenomics.com/csv/AMLBloodCell.csv>

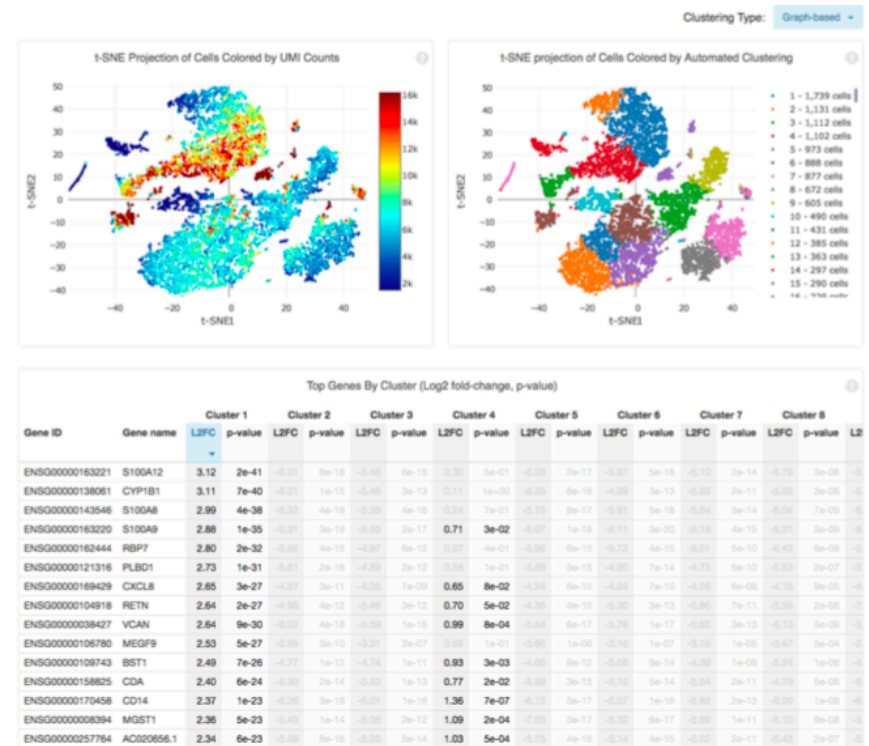
4. QC metrics: web_summary.html

Contains a “SUMMARY” and “ANALYSIS” tab:

The summary metrics describe sequencing quality and various characteristics of the detected cells.



Analysis tab contains results from automated secondary analysis



4. QC metrics: web_summary.html

We will go through the “SUMMARY” tab which provides QC metrics. Don’t worry about the “ANALYSIS” tab – we will look at the analysis results in Loupe Cell Browser.

Take your time to look at metrics for:

- Sequencing
- Mapping

Press the ? at the top right corner of the box for description of each reported metric.

We will look at the barcode rank plot in the “cells” box together.

4. QC metrics: [web_summary.html](#)

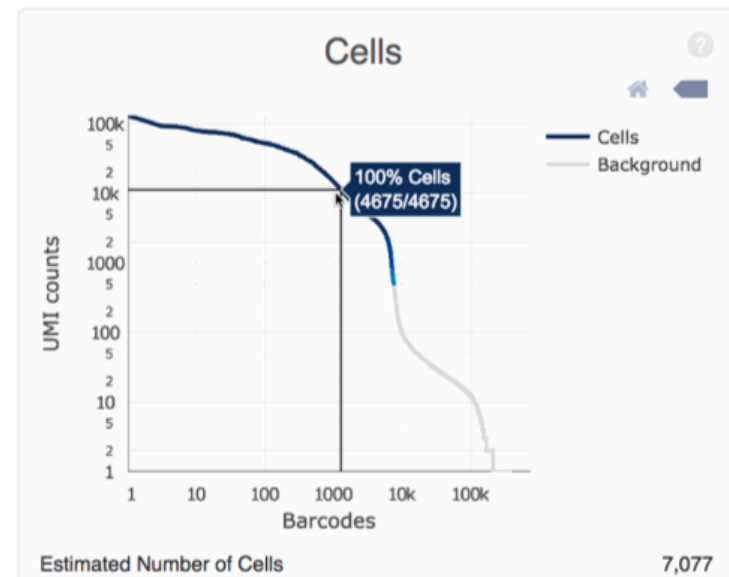
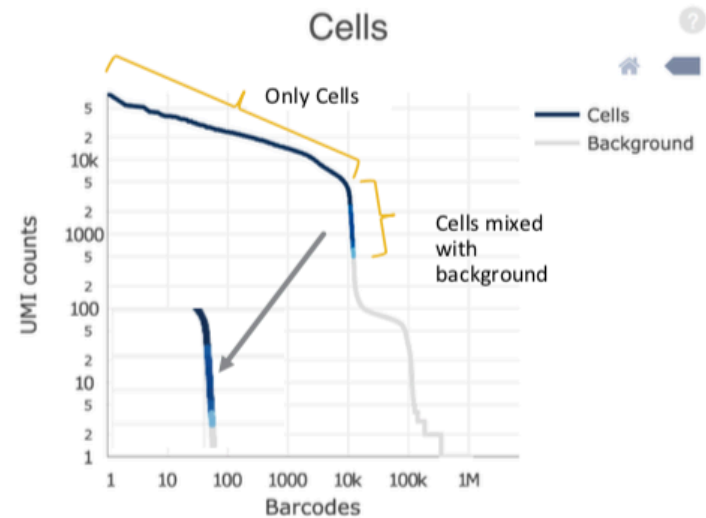
Barcode rank plot shows distribution of barcode counts and which barcodes were inferred to be associated with cells.

Cells are coloured in “blue”

Region on the plot where cells and background barcodes have similar UMI counts

Blue colour's gradient is proportional to the fraction of cells in a given subset of barcodes

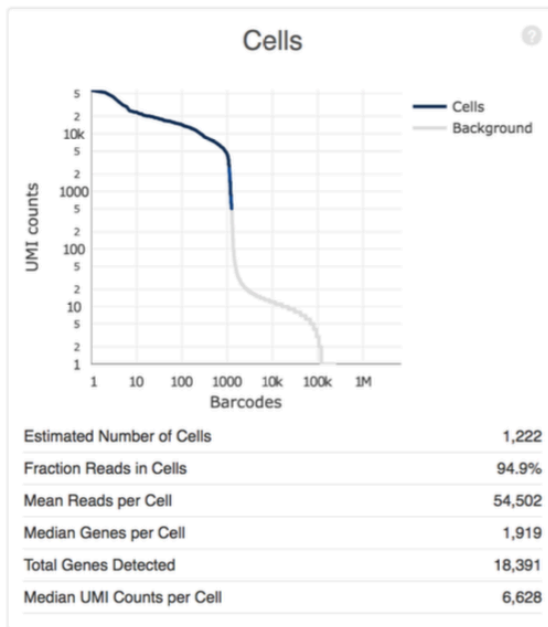
Tool tip that displays the fraction of cells versus total cells in a given region of the plot



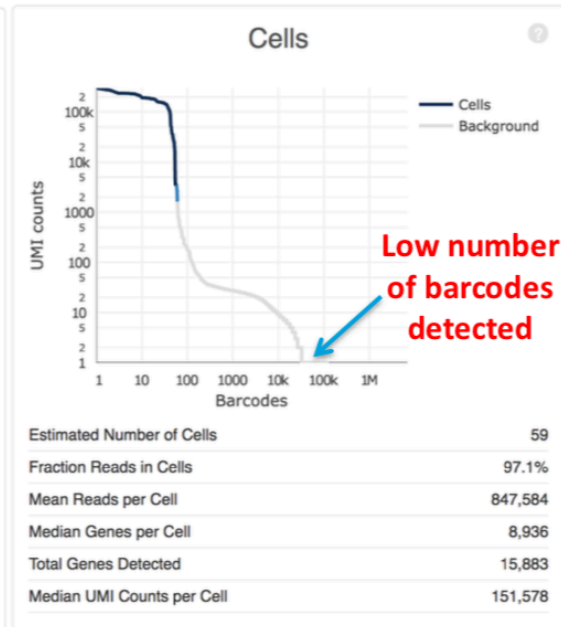
4. QC metrics: web_summary.html

Some examples of different quality runs:

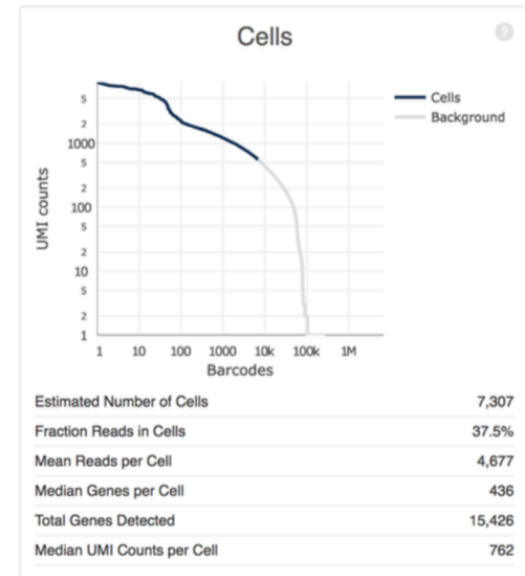
Typical Sample



Low Barcode counts
(eg. Clog, low-depth)



Loss of Single Cell Behavior
(eg. Wetting failure)



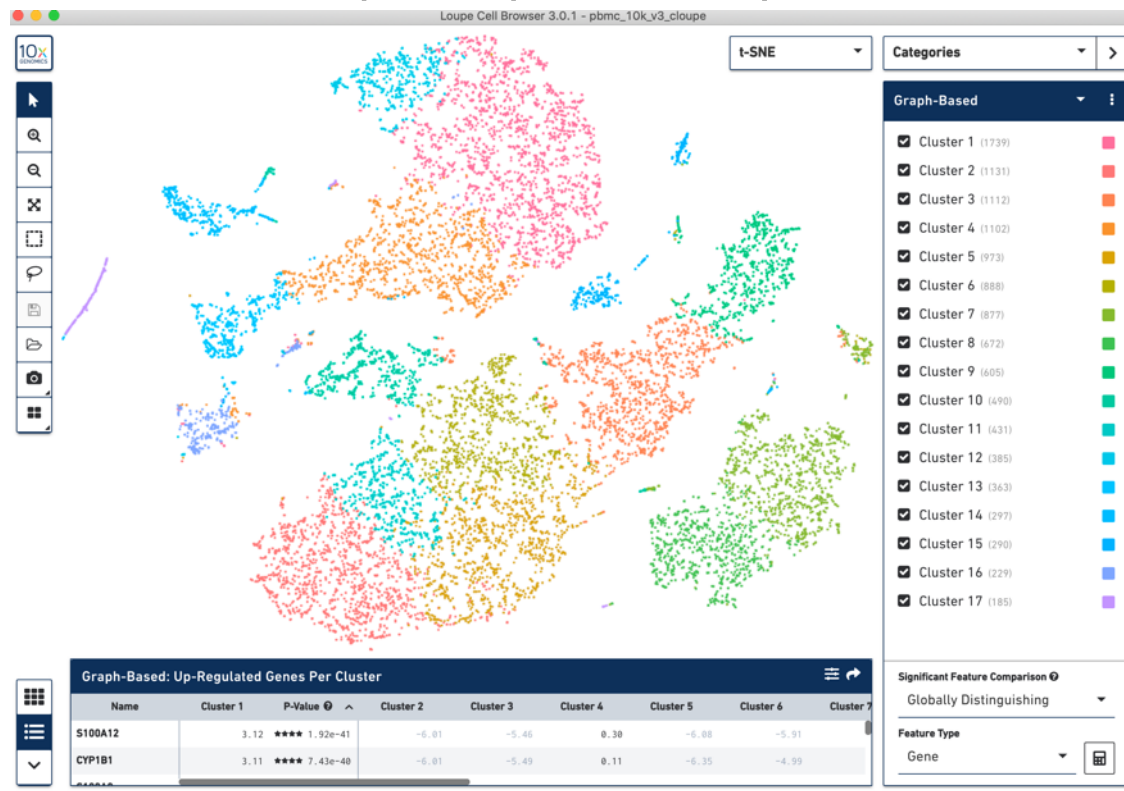
Algorithm has trouble discerning cells from the background

4. Loupe Cell Browser

Download the Loupe Cell Browser (10X Genomics):

<https://support.10xgenomics.com/single-cell-gene-expression/software/downloads/latest#loupe>

Open the pbmc_10k_v3_cloupe.cloupe file that you downloaded earlier.



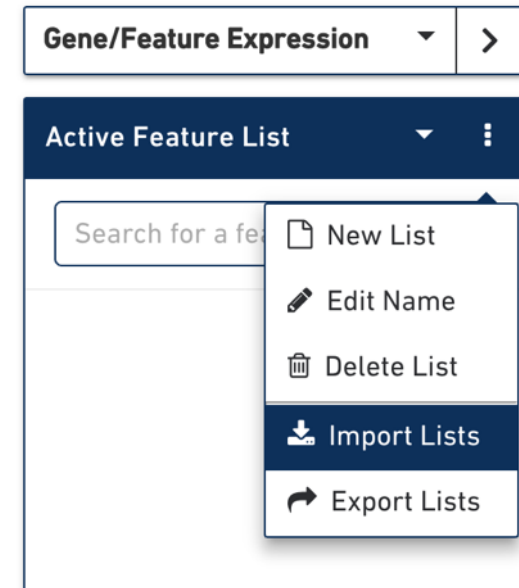
4. Loupe Cell Browser – Identify cell types

Explore:

- Graph-Based clustering results
- K-Means clustering results
- Differentially expressed genes per cluster
- Heatmaps

Identify cell types using 10X's gene list:

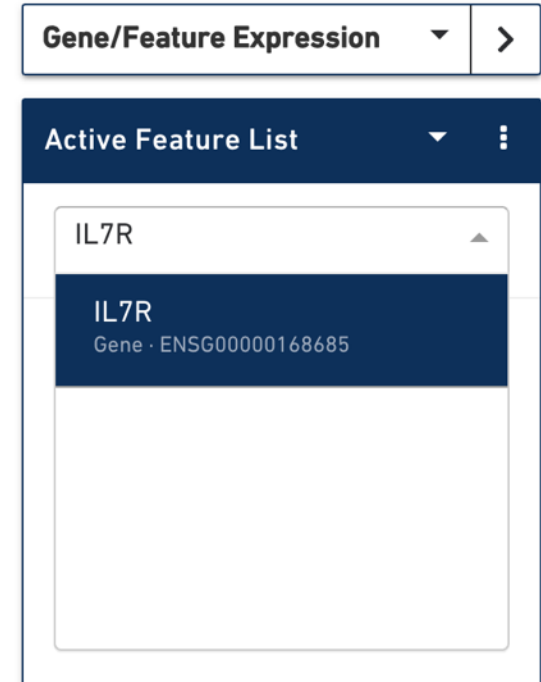
- Import “AMLBloodCell.csv” Gene list
- Identify the clusters most likely to be:
 - T Cells
 - Cytotoxic/CD8 T Cells
 - B cells
 - Monocytes
 - Proliferating Erythrocytes



4. Loupe Cell Browser – Identify cell types

Identify cell types using your own markers:

Cell type	Markers
CD4+ T Cells	IL7R
CD14+ Monocytes	CD14, LYZ
B cells	MS4A1
Dendritic cells	FCER1A, CST3
Megakaryocytes	PPBP



Section 5: Additional pipelines and resources



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5. Additional 10X Genomics pipelines

Pipeline	Functionality
cellranger mkfastq	Barcode-aware demultiplexing from BCL to FASTQ
cellranger count	• Read-level analysis of a single library • Produces feature/barcode matrix • Produces expression analysis and static visualizations • Produces .cloupe file for Loupe Cell Browser
cellranger aggr	• Aggregate `cellranger count` results • Combine data from multiple libraries and normalize depth • Optional chemistry correction for combining data between chemistries • Produces combined feature/barcode matrix • Produces combined expression analysis and static visualizations • Produces .cloupe file for Loupe Cell Browser
cellranger reanalyze	• Re-run the expression analysis with custom parameters • Produces .cloupe file for Loupe Cell Browser

5. Other resources

10X Chromium and Cell Ranger

- <https://www.10xgenomics.com/solutions/single-cell/>
- <https://support.10xgenomics.com/single-cell-gene-expression/software/overview/welcome>

Popular open source software:

- **Seurat:** <https://satijalab.org/seurat/>
- **Monocle2:** <http://cole-trapnell-lab.github.io/monocle-release/docs/>

Proprietary GUI software:

- **Partek Flow** (contact joey.lai@sydney.edu.au for access through the Westmead Medical Research Institute)